

VIJAY KUMAR R

AI Engineer | GenAI | LLM | Fintech | FRM

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PROFESSIONAL SUMMARY

AI Engineer with 2+ years of experience building production-grade Generative AI, LLM, RAG, NLP, and anomaly detection systems. Skilled in developing fintech AI solutions including FRM risk investigation assistants, OCR-based document automation, and real-time transaction monitoring. Strong hands-on experience with local LLM, prompt engineering, FastAPI, Gen AI, PySpark, Docker, and on-premise AI systems.

TECHNICAL SKILLS

AI & GenAI: Large Language Models (LLM · LLMs) · Generative AI (GenAI) · Retrieval-Augmented Generation (RAG) · Prompt Engineering · Embeddings · Vector Database · LangChain · FAISS · Vector Search · Ollama · LM Studio · HuggingFace Transformers
ML & NLP: Machine Learning (ML) · Deep Learning (DL) · Natural Language Processing (NLP) · Anomaly Detection · Optical Character Recognition (OCR) · Computer Vision · Speech Processing · Audio Feature Extraction (Librosa)
Frameworks: PyTorch · TensorFlow · Keras · Scikit-Learn · HuggingFace · OpenCV · NLTK · SpaCy · FastAPI
Data Engineering: Python · PySpark · Apache Kafka · Apache Airflow · PostgreSQL · Pandas · NumPy · SQL · REST APIs
Dev & Tools: Docker · Linux · Web Scraping · Selenium · Scrapy · Git
Fintech: Real-Time Transaction Monitoring · Anti-Money Laundering (AML) · Financial Risk Management (FRM) · Knowledge Graphs · Metabase

EXPERIENCE

AI Engineer

FIS - Financial Technology (via Futurz Staffing) | Apr 2025 - Present

- Built a real-time anomaly detection pipeline processing 50+ crore financial transactions, combining unsupervised Isolation Forest and supervised Random Forest models achieving 80%+ accuracy and significantly reducing false positive alerts in FRM workflows
- Developed an LLM-powered Risk Investigator Virtual Assistant for FRM case managers, enabling 360° customer profiling, transaction history analysis, and alert investigation through natural language prompts using local Qwen2.5 with quantization for optimized on-premise deployment.
- Engineered MailActra, an intelligent mail automation system for Law Enforcement Agency (LEA) department using OCR, local Qwen LLM and Generative AI to extract action items, classify mail intent and automate regulatory compliance workflows via Apache Kafka streaming

AI Engineer

Barrla Systems Pvt Ltd | Feb 2024 - Mar 2025

- Designed and implemented a custom NLP-based intent classification pipeline to identify food items, location entities and order intent from raw user prompts, briefly leading a team of 4 developers as Module Lead
- Built and evaluated multiple prototypes using open-source models including BERT, Mistral and custom Deep Learning and Machine Learning architectures to optimize intent prediction accuracy and model inference latency
- Conducted R&D on Small Language Models (SLMs), open-source LLMs and Generative AI techniques for on-device and low-resource AI agent deployment across conversational AI ordering workflows

KEY PROJECTS

1. Risk Investigator Virtual Assistant

Tech Stack: Python · LLMs · RAG Pipeline · Prompt Engineering · Qwen2.5 · Local LLM · Quantization · Conversation Memory · FastAPI · PostgreSQL · Docker · Linux

- Built an LLM-powered virtual assistant for FRM case managers enabling 360° customer profiling, real-time transaction analysis and alert investigation through natural language prompts using local Qwen2.5 with RAG pipeline, conversation memory and quantization deployed via FastAPI on Linux-based on-premise infrastructure

Highlights: Case Management · Fraud Pattern Recognition · Customer Profiling · Real-Time Transaction Monitoring · Financial Risk Management (FRM) · On-Premise Deployment · Agentic AI · Prompt Engineering

2. Anomaly Detection System - Financial Transactions

Tech Stack: Python · Isolation Forest · Random Forest · Machine Learning · Clustering · Feature Engineering · EDA · PySpark · Big Data · Parquet · SQL · On-Premise

- Engineered a real-time financial anomaly detection pipeline processing 50+ crore transactions using unsupervised Isolation Forest, supervised Random Forest and clustering techniques with feature engineering, EDA and big data handling via PySpark and Parquet achieving 80%+ accuracy and reducing false positive alerts

Highlights: Fraud Pattern Recognition · Financial Crime Detection · Real-Time Transaction Monitoring · False Positive Reduction · FRM · Big Data Analytics · On-Premise

3. MailActra - Intelligent Mail Automation System

Tech Stack: Python · LLMs · Generative AI · OCR · Prompt Engineering · Qwen2.5 · Local LLM · Quantization · Translation LLM · Apache Kafka · Docker

- Developed an end-to-end LEA department mail automation system using OCR-based document intelligence, local Qwen2.5 LLM with quantization and translation capabilities, Kafka-based streaming and Generative AI for automated action item extraction and regulatory compliance workflow management

Highlights: Document Intelligence · LEA Compliance · Workflow Automation · Regulatory Compliance · On-Premise Deployment · OCR · Apache Kafka · Generative AI

EDUCATION

B.Sc. Computer Science | ANJAC, Sivakasi, Tamil Nadu | 2023